

Lab Report 10

1. Objective

The objective of this lab is to understand call by value, call by reference, and default parameters in C++ functions.

2. Apparatus / Software

- Computer System
 - C++ Compiler (Code::Blocks / Dev C++ / VS Code)
-

3. Theory

Functions in C++ can be called in different ways. In call by value, a copy of the variable is passed. In call by reference, the actual variable is accessed. Default parameters allow functions to use predefined values when arguments are not provided.

4. Lab Tasks Implementation

Task 1: Call by Value (Swapping)

```
#include <iostream>
using namespace std;

// function jo values ko swap karti hai lekin original values change nahi
hoti
void swapping(int x, int y) {

    // temporary variable use karke swapping ki ja rahi hai
    int temp = x;
    x = y;
    y = temp;

    // function ke andar swapped values display ki ja rahi hain
    cout << "Function ke andar values: x = " << x << " y = " << y << endl;
```

```

}

int main() {

    int x = 50, y = 70;

    // original values display ki ja rahi hain
    cout << "Before function: x = " << x << " y = " << y << endl;

    // function call ki ja rahi hai
    swapping(x, y);

    // call by value ki wajah se original values change nahi hoti
    cout << "After function: x = " << x << " y = " << y << endl;

    return 0;
}

```

Task 2: Call by Reference (Swapping)

```

#include <iostream>
using namespace std;

// function jo reference ke zariye original values ko swap karti hai
void swapping(int &x, int &y) {

    // temporary variable use karke swapping ki ja rahi hai
    int temp = x;
    x = y;
    y = temp;
}

int main() {

    int x = 50, y = 70;

    // swapping se pehle values display ki ja rahi hain
    cout << "Before function: x = " << x << " y = " << y << endl;

    // function call ki ja rahi hai
    swapping(x, y);

    // swapping ke baad values change ho jati hain
    cout << "After function: x = " << x << " y = " << y << endl;

    return 0;
}

```

Task 3: Default Parameters

```

#include <iostream>
using namespace std;

```

```
// function jisme default values assign ki gayi hain
int divide(int a = 8, int b = 2) {

    // division perform ki ja rahi hai
    return a / b;
}

int main() {

    // koi argument pass nahi kiya gaya
    cout << divide() << endl;

    // sirf ek argument pass kiya gaya hai
    cout << divide(12) << endl;

    // dono arguments pass kiye gaye hain
    cout << divide(63, 7) << endl;

    return 0;
}
```

5. Conclusion

In this lab, call by value, call by reference, and default parameters were implemented successfully in C++ functions.